Q1- Write structures of different chain isomers of alkanes corresponding to the molecular formula C_6H_{14}. Also write their IUPAC names

Q2- Write structures of different isomeric alkyl groups corresponding to the molecular formula C5H11. Write IUPAC names of alcohols obtained by attachment of –OH groups at different carbons of the chain.

Q3- Write IUPAC names of the following compounds :
(i) $(CH_3)_3CCH_2C(CH_3)_3$
(ii) $(CH_3)_2C(C_2H_5)_2$
(iii) tetra – tert-butylmethane

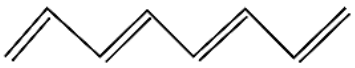
Q4- Write structural formulas of the following compounds :
(i) 3, 4, 4, 5–Tetramethylheptane
(ii) 2,5-Dimethyhexane

Q5- Write structures for each of the following compounds. Why are the given names incorrect? Write correct IUPAC names.
(i) 2-Ethylpentane
(ii) 5-Ethyl – 3-methylheptane

Q6- Sodium salt of which acid will be needed for the preparation of propane ? Write chemical equation for the reaction.

Q7- Write IUPAC names of the following compounds:

(i) $(CH_3)_2CH-CH=CH-CH_2-CH=CH-CH(C_2H_5)$

(ii) 

(iii) $CH_2=C(CH_2CH_2CH_3)_2$

(iv) $CH_3CH_2CH_2CH_2-CH=C(CH_2CH_3)-CH_2-CH(CH_3)$

Q8- Calculate number of sigma (σ) and pi (π) bonds in the above structures (i-iv).

Q9- Write structures and IUPAC names of different structural isomers of alkenes corresponding to C_5H_{10}

Q10- Draw cis and trans isomers of the following compounds. Also write their IUPAC names :
(i) CHCl = CHCl (ii) $C_2H_5CCH_3 = CCH_3C_2H_5$

Q11- Which of the following compounds will show cis-trans isomerism?
(i) $(CH_3)_2C = CH – C_2H_5$
(ii) $CH_2 = CBr_2$
(iii) $C_6H_5CH = CH – CH_3$
(iv) $CH_3CH = CCl\,CH_3$

Q12- Write IUPAC names of the products obtained by addition reactions of HBr to hex-1-ene

(i) in the absence of peroxide and (ii) in the presence of peroxide

Q13- Write structures of different isomers corresponding to the 5th member of alkyne series. Also write IUPAC names of all the isomers. What type of isomerism is exhibited by different pairs of isomers?

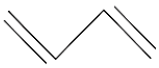
Q14- How will you convert ethanoic acid into benzene?

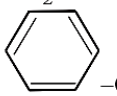
Q15- How do you account for the formation of ethane during chlorination of methane ?

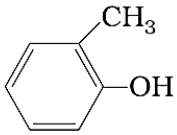
Q16- Write IUPAC names of the following compounds :

(a) $CH_3CH{=}C(CH_3)_2$

(b) $CH_2{=}CH{-}C{\equiv}C{-}CH_3$

(c) 

(d)  $-CH_2-CH_2-CH{=}CH_2$

(e) 

(f) $CH_3(CH_2)_4\,CH\,(CH_2)_3\,CH_3$
$\quad\quad\quad

Q29- What effect does branching of an alkane chain has on its boiling point?

Q30- Addition of HBr to propene yields 2-bromopropane, while in the presence of benzoyl peroxide, the same reaction yields 1-bromopropane. Explain and give mechanism.

Q31- Write down the products of ozonolysis of 1,2-dimethylbenzene (o-xylene). How does the result support Kekulé structure for benzene?

Q32- Arrange benzene, n-hexane and ethyne in decreasing order of acidic behaviour. Also give reason for this behaviour.

Q33- Why does benzene undergo electrophilic substitution reactions easily and nucleophilic substitutions with difficulty?

Q34- How would you convert the following compounds into benzene? (i) Ethyne (ii) Ethene (iii) Hexane

Q35- Write structures of all the alkenes which on hydrogenation give 2-methylbutane.

Q36- Arrange the following set of compounds in order of their decreasing relative reactivity with an electrophile, E+

(a) Chlorobenzene, 2,4-dinitrochlorobenzene, p-nitrochlorobenzene

(b) Toluene, p-$H_3C - C_6H_4 - NO_2$, p-$O_2N - C_6H_4 - NO_2$.

Q37- Out of benzene, m–dinitrobenzene and toluene which will undergo nitration most easily and why?

Q38- Suggest the name of a Lewis acid other than anhydrous aluminium chloride which can be used during ethylation of benzene.

Q39- Why is Wurtz reaction not preferred for the preparation of alkanes containing odd number of carbon atoms? Illustrate your answer by taking one example.